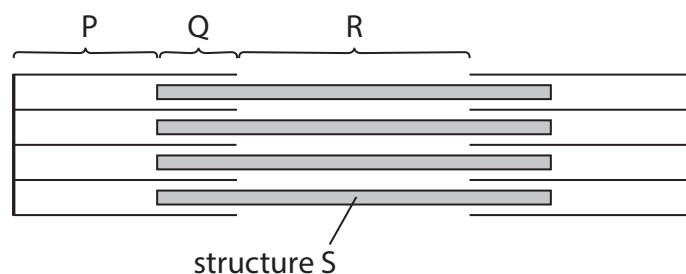


4 The sarcomere is the contractile unit of skeletal muscle fibres.

(a) The diagram shows a relaxed sarcomere.



(i) Which of the following would shorten when the sarcomere contracts?

(1)

- ☐ **A** region P only
- ☐ **B** region R only
- ☐ **C** regions P and R
- ☐ **D** regions Q and R

(ii) A diagram of this sarcomere, drawn to the same scale, has a length of 50 mm when contracted.

Give the ratio of the lengths of the relaxed sarcomere to the contracted sarcomere.

(1)

Answer : 1

(iii) Name the major protein component that forms structure S in the diagram.

(1)

.....

(b) Describe the role of troponin molecules in the contraction of a sarcomere.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

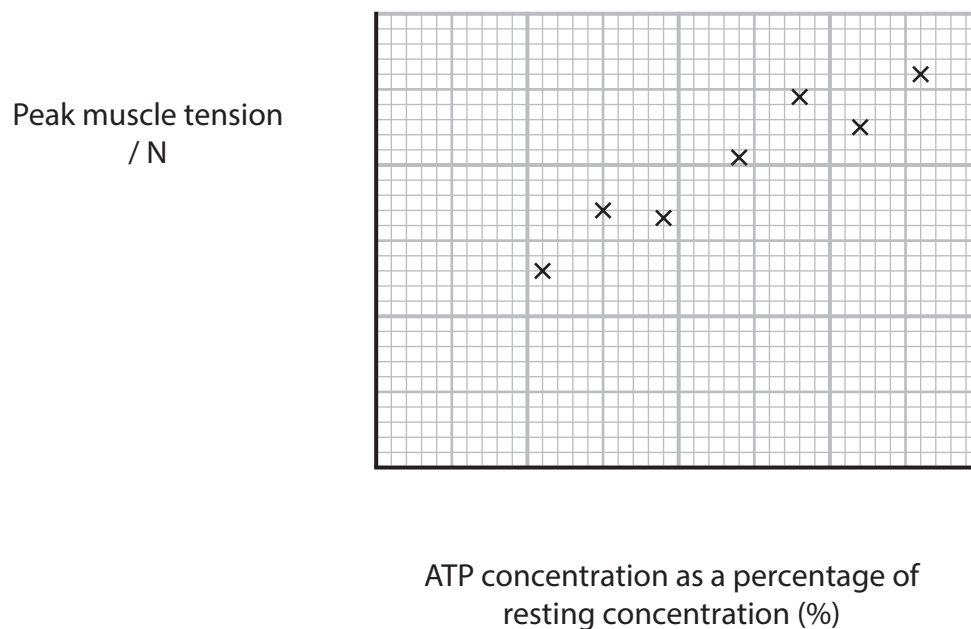
(c) Muscle contraction increases muscle tension.

The relationship between increasing ATP concentration and peak muscle tension was investigated.

The results are shown in the table.

ATP concentration as a percentage of resting concentration (%)	Peak muscle tension / N
110	26
150	34
190	33
240	41
280	49
320	45
360	52

The incomplete graph shows the results of this investigation.



(i) Add an appropriate scale to the axes of the graph and draw a line of best fit.

(2)

(ii) Calculate the gradient of the line of best fit.

(1)

Answer N per percentage increase

(Total for Question 4 = 10 marks)
